HSE Intelligence Platform

Formulas & Data Source Reference

Every chart, KPI, gauge and percentage across the application: the exact calculation, which database table/column it reads (and the originating HSE_Intelligence_Test_Data.xlsx sheet where applicable), and whether it is a real live calculation, a documented best-effort proxy, a known bug, or static placeholder data.

Legend:

REAL	REAL - live DB calculation
PROXY	BEST-EFFORT PROXY - real data, approximated concept
BUG	KNOWN BUG - today-anchored date window reads 0 on this dataset
DUMMY	STATIC DUMMY - no backing table exists

Contents

Dashboard (Home)

Route: /

Predictive Injury Risk Score

[BUG]

Source: incidents.severity, incidents.incident_date_time

Formula:

```
weight: Critical/Significant=3, High/Major=2, Medium/Moderate=1, else=0.5
score(period) = min(100, SUM(weight) / (count x 3) x 100)
current  = score(today-90d -> today)
previous = score(today-180d -> today-90d)
Score = round(current); Trend = round(current - previous)
```

Notes: *backend/app/controllers/dashboard.py:get_leading_indicators(). Anchored to real today (2026); dataset activity ends 2025, so this currently reads 0%.*

TRIR / LTIF

[PROXY]

Source: incidents.incident_date_time, incidents.days_away, employees (count)

Formula:

```
hours_worked = total_employees x 2,000 (assumed annual hrs/employee)
recordable = incidents in trailing 12 months
lost_time  = incidents in trailing 12 months WHERE days_away > 0
TRIR = recordable x 200,000 / hours_worked
LTIF = lost_time  x 1,000,000 / hours_worked
```

Notes: *Same file. Real-time formula; not a certified OSHA-audited figure since actual hours-worked isn't tracked in this schema.*

Contractor Risk Score

[REAL]

Source: employees.employment_type, incidents.reported_by

Formula:

```
contractor_rate = contractor_incidents / contractor_employee_count
permanent_rate  = permanent_incidents  / permanent_employee_count
relative_risk = contractor_rate / permanent_rate
Score = round(min(100, relative_risk x 50))
Label: High >=1.5, Medium >=1, else Low
```

Notes: *Same file. employment_type matched case-insensitively on "contract".*

Audit Readiness Score

[BUG]

Source: safety_walks.compliance_rating

Formula:

```
Score = round(avg(compliance_rating over trailing 90 days) / 5 x 100)
```

```
Label: Ready >=80, Needs Attention >=60, else Not Ready
```

Notes: *Same file. Anchored to real today; reads 0%/Not Ready for the same reason as Predictive Injury Risk Score above.*

Top Risk Chart (Data-Based)

[PROXY]

Source: hazard_categories, hazards, incidents (joined via hazard_id)

Formula:

```
"data" series = count of incidents per hazard category (top 8)
"intelligence" series = max(0, count - 5)
```

Notes: *backend/app/controllers/dashboard.py:get_incidents_by_category(). The "intelligence" series is an arbitrary offset, not a separate real metric.*

Exposure Index (gauge)

[REAL]

Source: safety_walks.compliance_rating (all-time avg)

Formula:

```
value = round(avg_compliance_rating x 20)
```

Notes: *backend/app/controllers/dashboard.py:get_dashboard_stats() -> avg_compliance_rating.*

Competency Coverage (gauge)

[REAL]

Source: capa_actions (completed vs total)

Formula:

```
value = round(capa_completed / capa_total x 100)
```

Notes: *Same endpoint -> capa_completion_rate.*

Ranked Action Table / Overdue CAPA

[REAL]

Source: capa_actions, employees

Formula:

```
Priority: High if overdue, Medium if due in <=7 days, else Low.
Overdue list: status != Completed AND due_date < today.
```

Notes: *backend/app/controllers/dashboard.py:get_ranked_capa_actions() / get_overdue_capa().*

People (Users page, top section)

Route: /users

Competency Coverage % (+ sparkline)

[REAL]

Source: incidents.root_cause / root_cause_category, capa_actions.root_cause_addressed, employees.induction_date

Formula:

```
flagged = employees whose incident root_cause/category OR CAPA root_cause_addressed
contains "train"
Coverage = (total_employees - len(flagged)) / total_employees x 100
Sparkline: same formula recomputed at 10 monthly checkpoints as flags accrue by real
incident date
```

Notes: *backend/app/controllers/people.py:get_people_overview(). induction_date is populated for all 150 employees post-Excel-seed.*

Worker Exposure Index (gauge)

[REAL]

Source: incidents + near_misses, employees (count)

Formula:

```
Index = round(min(100, (recent_incidents + recent_near_misses) / total_employees x 100))
"recent" = trailing 90 days, anchored to the LATEST real incident/near-miss date in the
data (not today)
Label: High Risk >30, Medium Risk >=10, else Low Risk
```

Notes: *Same file. Anchoring fixed to use real data's own latest date.*

Supervisor Safety Score (ring)

[REAL]

Source: safety_walks.compliance_rating, roles.safety_signatory

Formula:

```
supervisors = employees whose role.safety_signatory = 'Yes'
Score = round(avg(compliance_rating of walks performed by supervisors) / 5 x 100)
```

Notes: *Same file.*

Fatigue Risk (overtime vs normal hours)

[REAL]

Source: shift_schedule.actual_hours_worked

Formula:

```
Per shift: normal = min(hours, 8); overtime = max(0, hours - 8)
Summed org-wide per week, over the most recent 10 weeks of real shift data
```

Notes: *Same file. NOTE: every one of the 73,220 real shifts logs exactly 8.5 hours with zero variance, so this is mathematically flat (5,600 normal / 350 overtime every week) -- that's the real data, not a bug.*

Safety Toolbox Meetings Trend

[PROXY]

Source: safety_walks.inspection_date_time (ALL types)

Formula:

```
Monthly count of every safety_walks row, anchored to the latest real safety_walk date
```

Notes: *Same file. PROXY: no "Toolbox" inspection_type exists anywhere in the Excel/DB (only Compliance / Follow-up / Routine) -- this counts ALL inspections, not real toolbox talks.*

High Risk Roles

[REAL]

Source: incidents.reported_by + near_misses.reported_by, employees.role_id, roles

Formula:

```
rate = (incidents + near_misses attributed to a role) / headcount in that role
Label: High if rate>=3, Medium if rate>=1.5, else Low. Top 4 shown.
```

Notes: *Same file.*

Training Expiry Status

[PROXY]

Source: employees.induction_date, training_programs.expiry_months

Formula:

```
For every (employee, training program) pair:
  next_due = induction_date, advanced by expiry_months until >= today
  bucket by days_until_due: Expired <0, Due<30 Days, Due<90 Days
Expiring Soon badge = Expired_count + Due<30_count
```

Notes: *Same file. HEURISTIC: assumes every employee must renew every catalog training on its own cycle -- there is no real per-employee training assignment table in this schema.*

Behaviour Observations

[REAL]

Source: safety_walks.issues_found, near_misses (count)

Formula:

```
Safe = safety_walks with issues_found = 0
At-Risk = safety_walks with issues_found > 0
Near Miss = count(near_misses)
Each shown as % of (Safe + At-Risk + Near Miss)
```

Notes: *Same file.*

Coaching Actions / Open Actions

[REAL]

Source: capa_actions.action_type, status, due_date

Formula:

```
Coaching = open CAPA where action_type = 'Training'
Open Actions = open CAPA where action_type != 'Training', ordered by due_date
```

Notes: *Same file. Coaching list is currently empty -- 0 of 42 real CAPA rows have action_type='Training' (all are 'Corrective').*

Employee Directory table

[REAL]

Source: employees JOIN roles JOIN departments JOIN sites

Formula:

```
Plain listing, no calculation -- real name/role/department/site/status per employee.
```

Notes: *backend/app/controllers/people.py:get_employee_directory().*

Root Cause Analysis

Route: /root-cause-analysis

Priority

[REAL]

Source: incidents.severity

Formula:

```
Significant / Critical / Fatal -> Critical
Lost Time / High / Major -> High
Serious / Medium / Moderate -> Medium
else -> Low
```

Notes: *backend/app/controllers/analytics.py:_rca_priority(). Tuned to this dataset's real severity vocabulary (Lost Time/Minor/Serious/Significant).*

Status

[REAL]

Source: incidents.investigation_status

Formula:

```
contains "complete" -> Closed; contains "progress" -> In Progress; else -> Pending
```

Notes: *analytics.py:_rca_status().*

Completion Date

[REAL]

Source: capa_actions.due_date (for that incident's CAPA rows)

Formula:

```
If status=Closed: use MAX(due_date) of CAPA marked 'Completed';
else fall back to MAX(due_date) of ANY CAPA tied to the incident;
else leave blank (no real data to derive it from)
```

Notes: *analytics.py:get_root_cause_analysis(). 10 of 36 Closed incidents have no CAPA at all, so stay blank -- that's a real data gap, not a bug.*

Corrective Actions / Preventive Measures

[REAL]

Source: capa_actions.description, action_type

Formula:

```
Preventive = CAPA descriptions where action_type contains "prevent"
Corrective = all other CAPA descriptions for that incident, joined with "; "
```

Notes: *Same file. Preventive is always "--" -- 0 of 42 real CAPA rows have a "preventive"-type action_type (all are 'Corrective').*

Site / Zone / Conducted By

[REAL]

Source: working_stations.zone_classification, sites.site_name, employees.full_name

Formula:

```
Site = site name via incident -> station -> site
Zone = station's zone_classification
Conducted By = the employee who reported the incident (closest real proxy; there's no dedicated "investigator" field)
```

Notes: *Same file.*

Actions / Work

Route: /actions

Active Permits

[REAL]

Source: permits_to_work.status

Formula:

```
count WHERE status = 'Active'
```

Notes: *backend/app/controllers/analytics.py:get_permits_summary().*

Work Exposure Hours

[REAL]

Source: permits_to_work.duration_requested_hours, number_of_workers

Formula:

```
SUM(duration_requested_hours x number_of_workers) WHERE status = 'Active'
```

Notes: *Same file.*

Permit Compliance %

[REAL]

Source: permits_to_work.deviation_reported, incident_occurred

Formula:

```
compliant = permits WHERE deviation_reported != 'Yes' AND incident_occurred != 'Yes'
Compliance % = compliant / total x 100
```

Notes: *Same file. Also reused identically on the Compliance page.*

High Risk Work (radar)

[REAL]

Source: permit_types JOIN permits_to_work (status='Active')

Formula:

```
count of active permits per permit type, normalized:
value = round(count / max(count across all types) x 100)
```

Notes: *Same file.*

Work by Permit Type (stacked bar)

[PROXY]

Source: permits_to_work.status, grouped by permit_type

Formula:

```
For each permit type: % of its permits that are Active / Closed / Expired (sums to ~100%
per type)
```

Notes: *Same file. REPLACES a fake "Work by Contractor" chart -- no contractor entity exists anywhere in this schema.*

Missing Work Controls

[REAL]

Source: permits_to_work.deviation_reported='Yes', status='Active'

Formula:

```
Real active permits with a reported deviation, soonest-expiring first
```

Notes: *Same file. REPLACES 4 fully fake static strings.*

Permit Violations

[REAL]

Source: incidents.permit_active='Yes' JOIN working_stations

Formula:

```
Recent incidents where a permit was marked active at the time
```

Notes: *Same file.*

Active Work Table / Expiry Timeline

[REAL]

Source: permits_to_work (status='Active'), validity_end

Formula:

```
"Expiry" column shows the REAL validity_end date/time -- not a countdown.
```

Notes: *Same file. Checked: only 2 of 828 'Active' permits are genuinely within their own validity window at any single timestamp, so a "time remaining" countdown can't be made honest for this dataset.*

Compliance

Route: /compliance

Compliance Score

[REAL]

Source: (composite of the 3 scores below)

Formula:

```
round(avg(Permit Compliance %, Legal Register Coverage %, Audit Readiness %))
```

Notes: *backend/app/controllers/analytics.py:get_compliance_summary().*

Legal Register Coverage %

[PROXY]

Source: policies.category (distinct), hazard_categories.category_name (distinct)

Formula:

```
min(100, round(distinct_policy_categories / distinct_hazard_categories x 100))
```

Notes: *Same file. Coverage-breadth proxy -- categories aren't named identically between the two registers (12 policy categories / 10 hazard categories).*

Audit Readiness Score

[REAL]

Source: safety_walks.compliance_rating (ALL-TIME avg)

Formula:

```
round(avg(compliance_rating) / 5 x 100)
```

Notes: *Same file. Deliberately all-time (not a trailing window) to avoid the today-anchored bug seen on the Dashboard page.*

Compliance Trend (+ MoM badge)

[REAL]

Source: safety_walks.compliance_rating, grouped by month

Formula:

```
Monthly: round(avg(compliance_rating)/5 x 100), last 10 months of real data, anchored to
the dataset's own latest safety_walk date
MoM = latest_month_value - prior_month_value
```

Notes: *Same file.*

Audit Findings by Severity

[PROXY]

Source: safety_walks WHERE inspection_type='Compliance'

Formula:

```
Critical: critical_issues >= 2
Major: critical_issues == 1
Minor: critical_issues == 0 AND issues_found >= 1
```

```
Observation: critical_issues == 0 AND issues_found == 0
```

Notes: *Same file. "Compliance" is the closest real proxy to an "audit" inspection type in this dataset.*

Permit Compliance (gauge)

[REAL]

Source: permits_to_work

Formula:

```
Identical formula to the Actions page.
```

Notes: *Same file.*

Policy Review Status (gauge)

[REAL]

Source: policies.status

Formula:

```
% of policies WHERE status = 'Current'
```

Notes: *Same file. Currently 100% -- all 12 real policies are marked Current.*

Non-Conformance table

[REAL]

Source: capa_actions (open) + incidents.severity

Formula:

```
Open CAPA actions; criticality = the linked incident's Priority (Critical/High->High,
Medium->Medium, Low->Low)
```

Notes: *Same file.*

Analytics & Reports

Route: /analytics

Violation Type Breakdown (pie)

[REAL]

Source: incidents.root_cause_category

Formula:

```
count per category, top 5 (this dataset has exactly 5: Equipment, Management System, Procedure, Supervision, Training)
```

Notes: *backend/app/controllers/analytics.py:get_violations_summary().*

Zone Risk Distribution (bar)

[REAL]

Source: incidents JOIN working_stations

Formula:

```
count of incidents per station, top 7 -- then on THIS page, normalized to 0-100% relative to the busiest station before applying red/amber/green color thresholds
```

Notes: *Backend returns raw counts (shared with Violations page); AnalyticsPage.tsx normalizes locally. Fixed: raw counts (max ~4) never crossed the 40/70 thresholds, so every bar used to render green regardless of real risk.*

Monthly Violations vs Near Misses

[REAL]

Source: incidents.incident_date_time, near_misses.event_date_time

Formula:

```
count per month, grouped, last N months
```

Notes: *Same backend endpoint.*

Severity Mix (Violations page)

[REAL]

Source: incidents.severity, grouped by month

Formula:

```
Same Critical/High/Medium/Low mapping as Root Cause Analysis Priority (fixed -- previously used a different, incorrect keyword mapping)
```

Notes: *analytics.py:get_violations_summary(), now reuses _rca_priority().*

Downtime by Type

[REAL]

Source: incidents.days_away, summed per incident_type

Formula:

```
SUM(days_away) per type, the 5 LOWEST-downtime types are shown (ascending order)
```

Notes: *Same endpoint. Worth reviewing: typically you'd want the highest-downtime types surfaced, not the lowest.*

PPE Compliance tab

[DUMMY]

Source: (none)

Formula:

```
Hard Hat 96%, Vest 92%, Shoes 89%, Gloves 85%, Goggles 78%
```

Notes: *FrontendNew/src/app/pages/AnalyticsPage.tsx. DUMMY DATA kept by explicit request (2026-06-17) -- no PPE-tracking table exists anywhere in the schema.*

Scheduled Reports table

[DUMMY]

Source: (none)

Formula:

```
3 fake report rows with fake emails/dates
```

Notes: *Same file. DUMMY DATA kept by explicit request (2026-06-17) -- no report-scheduling/email table exists in the schema.*

Sites checklist (Custom Reports tab)

[REAL]

Source: sites.site_name

Formula:

```
Real site names from GET /sites/
```

Notes: *Same file. NOTE: source Excel genuinely has all 8 Site rows duplicated as "Bridgend Manufacturing Complex" -- that's a real characteristic of the source data, not a bug.*

Risk Management

Route: /risk

Control Effectiveness Score

[REAL]

Source: capa_actions

Formula:

```
round(completed / total x 100)
```

Notes: *backend/app/controllers/analytics.py:get_risk_summary().*

Unverified Controls

[REAL]

Source: capa_actions.status != 'Completed'

Formula:

```
count
```

Notes: *Same file.*

Risk Escalations

[REAL]

Source: capa_actions.status != 'Completed' AND due_date < today

Formula:

```
count
```

Notes: *Same file.*

Risk by Zone/Site/Team (bar)

[REAL]

Source: incidents JOIN working_stations JOIN sites

Formula:

```
count of incidents per site, top 5
```

Notes: *Same file.*

Action/High Risk Active Tasks table

[REAL]

Source: capa_actions (open), employees

Formula:

```
Status label: Overdue (Red) if due_date<today; In Progress (Amber) if due in <=3 days;
else Pending (Yellow)
```

Notes: *Same file.*

Risk Aging (stacked bars + line)

[REAL]

Source: capa_actions (open)

Formula:

```
Bucket by days overdue: 0-30 / 31-60 / 61-90 / >90 Days
Sub-severity within bucket by days_over: >60->critical, >30->high, >0->medium, else low
(no due_date -> medium)
```

Notes: *Same file.*

Residual Risk Trend (area chart)

[DUMMY]

Source: (none)

Formula:

```
Q1 90, Q2 64, Q3 48, Q4 34
```

Notes: *FrontendNew/src/app/pages/RiskPage.tsx. Fully static -- no quarterly risk model exists in the backend.*

Risk Matrix (5x5 heatmap)

[DUMMY]

Source: (none)

Formula:

```
Static likelihood x impact scores/labels
```

Notes: *Same file. Fully static reference matrix, not backend-driven.*

Violations

Route: /violations

Incident Types / Incident Location

[REAL]

Source: incidents.incident_type, working_stations.station_name

Formula:

```
Same by_type / by_location formulas as the Analytics page (shared endpoint)
```

Notes: *backend/app/controllers/analytics.py:get_violations_summary().*

Injury Category / Person Involved / Type of Injury

[DUMMY]

Source: (none)

Formula:

```
e.g. Hand/Finger 5, Multiple 3... / Green Hand(s) 6... / Cut Wound 3...
```

Notes: *FrontendNew/src/app/pages/ViolationsPage.tsx. Fully static -- no injury-body-part or person-type tracking exists in this schema.*

Incident Cause Category (pie)

[REAL]

Source: incidents.root_cause_category

Formula:

```
Top 3 categories by count (cause_data)
```

Notes: *Shared endpoint.*

Incident Trend / Near Miss Trend / Downtime / Severity Mix / RCA Breakdown

[REAL]

Source: incidents, near_misses (shared with Analytics/Dashboard)

Formula:

```
Same formulas documented under Analytics & Reports above.
```

Notes: *Shared endpoint.*

Key Learnings

[DUMMY]

Source: (none)

Formula:

```
4 static placeholder strings
```

Notes: *Same file. Fully static, and the placeholder text itself appears to be garbled filler, not even meaningful dummy content.*

Open Actions checklist

[REAL]

Source: capa_actions (open)

Formula:

```
Same open_capa_items list as Dashboard/Analytics
```

Notes: *Shared endpoint.*

Assets

Route: /equipment (or similar)

Asset Control Effectiveness, Safety Maintenance Status, Critical Asset Failure Trends, Asset Heat Map, Asset Incident Trend, Asset Risk Table, Overdue Inspections, Critical Barrier Checklist

[DUMMY]

Source: (none)

Formula:

```
All hardcoded numbers/labels.
```

Notes: *FrontendNew/src/app/pages/AssetsPage.tsx. ENTIRE PAGE is fully static -- there is no equipment/asset table anywhere in this schema.*